5 February 1973
SUPERSEDING
MIL-B-20100B
10 October 1968

MILITARY SPECIFICATION

BRUSHES, CLEANING, FOR BORE, CHAMBER AND FIRING PIN HOLE OF SMALL ARMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers bore, chamber and firing pin hole cleaning brushes for small arms.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military		
MIL-P-116	-	Preservation, Methods of.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of.

STANDARDS

Military		
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

* 3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include the pilot pack (see 5.1).

* 3.2 Materials, construction and design. Brushes shall conform to the materials, construction and design requirements in accordance with the applicable drawings specified in the contract (see 6.2) and MIL-W-13855.

* 3.2.1 Bristle weight. The bristle weight shall be in accordance with the applicable drawings specified in the contract (see 6.2).

* 3.2.2 Bristle distribution. Unless otherwise specified on the drawings, bristles shall be evenly distributed.

* 3.2.3 Protective finish. The protective finish shall be in accordance with the applicable drawings specified in the contract (see 6.2).

* 3.2.4 Nail load. The nail load, for ratchet type brushes only, shall be in accordance with the applicable drawings specified in the contract (see 6.2).

* 3.2.5 Tensile pull. The tensile pull, as applicable, shall be in accordance with the applicable drawings specified in the contract (see 6.2).

* 3.2.6 Ratchet type brushes. Ratchet type brushes, after final assembly, shall have freely rotating ratchet members.

3.3 Performance.

3.3.1 Bore cleaning brushes. Brushes shall withstand 1000 strokes through the full length of the type and caliber barrel specified. After 1000 strokes, the bristles shall not be loose in the core wire and bristle weight shall not be less than 75 percent of the original weight.

* 3.3.2 Chamber cleaning brushes. Brushes shall withstand 1000 insertions into the full depth of the applicable chamber and while fully inserted on each insertion, the brush shall be rotated one (1) full revolution. Speed of revolution shall be 15 to 30 revolutions per minute. After 1000 revolutions and insertions, the bristles shall not be loose in the core wire and bristle weight shall not be less than 75 percent of the original weight.

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3.3.3 Firing pin hole cleaning brush. Brushes shall withstand 1000 strokes into the specified firing pin hole of the applicable bolt. After 1000 strokes, the bristles shall not be loose in the core wire and bristle weight shall not be less than 75 percent of the original weight.

* 3.4 Workmanship. Cleaning brushes shall be in accordance with the workmanship requirements of MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

* 4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. A first article shall be selected from early production and submitted for testing in accordance with contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

* 4.5.1 Inspection lot. The formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be no larger than one week's production and within the limitations of MIL-W-13855.

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4.5.2 Sampling.

4.5.2.1 Sampling for examination of brushes shall be in accordance with MIL-STD-105 inspection level I.

4.5.2.2 Sampling for packaging shall be in accordance with MIL-P-116.

4.5.2.3 Sampling for packing shall be in accordance with MIL-STD-105 using inspection level I.

4.5.2.4 Sampling for the bristle weight test shall be in accordance with inspection level S2 of MIL-STD-105. Acceptance shall be based on an AQL of 1.0 percent defective.

4.5.2.5 The sample for the performance, nail load and tensile pull tests as applicable shall be ten brushes randomly selected from each lot.

4.5.3. Examination.

4.5.3.1 Brushes. Examination of brushes shall be performed in accordance with the applicable inspection provisions of MIL-W-13855 and the following:

a. Each brush of the sample shall be inspected for the applicable defects listed in Table I.
b. The AQL listed shall be applied to the group of defects, not to an individual defect.
c. Where visual is specified as the method of inspection, the sample shall be either scaled or compared with an accepted sample.
* d. Where manual is specified as the method of inspection, the sample shall be examined for secureness of assembly of the bristles in the core and, as applicable, free rotation between ratchet members.
e. Where SME (Standard Measuring Equipment) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection equipment approved by the Government.
f. Pieces examined for workmanship shall be carefully compared with a specimen of known acceptable quality, provided by the contracting officer, and shall conform to the requirements of 3.4.

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TABLE I

Classification of Defects

Categories	Defect	Inspection Method
Critical:	None defined.	
Major:	AQL = 2.5 percent defective	
101	Brush diameter	SME
102	Thread characteristics	SME
103	Improper assembly or missing parts	Visual
* 104	Improper bristle distribution	Visual
105	Bristles not secure	Manual
* 106	Ratchet members do not rotate freely	Manual
Minor:	AQL = 6.5 percent defective	
201	Eccentricity of thread and tip	SME
202	Core (wire) diameter	SME
203	Presence and continuity of protective coating	Visual
204	Overall length	Visual
205	Length of bristle portion	Visual
206	Pitch or number of turns of assembled brush	Visual
207	Workmanship	Visual

* 4.5.3.2 Packaging. Examination of packaging shall be as prescribed for group A and B of MIL-P-116.

4.5.3.3 Packing. Examination of packing shall be in accordance with MIL-P-14232.

4.5.4 Testing.

4.5.4.1 Government furnished property. Barrels and bolts to be used for the applicable performance test shall be of the type and caliber specified on the applicable brush drawing. Each barrel and bolt shall be clean and dry and no barrel or bolt shall be used for more than 500 brush tests.

4.5.4.2 Bristle weight. Brushes selected in accordance with 4.5.2.4 shall be tested in accordance with 4.6.1 for compliance with the bristle weight specified on the applicable drawing.

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4.5.4.3 Performance-bore cleaning brushes. Brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.2 for compliance with 3.3.1. Failure of any brush shall be cause for rejection of the represented lot.

4.5.4.4 Performance-chamber cleaning brushes. Brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.3 for compliance with 3.3.2. Failure of any brush shall be cause for rejection of the represented lot.

4.5.4.5 Performance-firing pin hole cleaning brushes. Brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.4 for compliance with 3.3.3. Failure of any brush shall be cause for rejection of the represented lot.

4.5.4.6 Nail load. When applicable, brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.5 for compliance with the nail load requirements specified on the applicable drawing. Failure of any brush shall be cause for rejection of the represented lot.

* 4.5.4.7 Tensile pull. When applicable, brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.6 for compliance with the tensile pull requirements specified on the applicable drawing. Failure of any brush shall be cause for rejection of the represented lot.

4.6 Test methods.

4.6.1 Bristle weight. The bristles of each brush shall be removed from the brush and weighed. The brush shall have failed when the bristle weight is not within the limits of the bristle weight specified in the applicable brush drawing.

4.6.2 Performance-bore cleaning brushes. The brush shall be pushed in one continuous motion through the length of the applicable barrel and returned. This movement shall be counted as two strokes. The brush shall be forced entirely clear of the barrel at the end of each stroke. Speed of travel for each stroke shall be approximately 2 feet per second. The number of strokes required for each test brush shall be as specified in 3.3.1. Barrels used for this test shall be of the type and caliber specified on the applicable brush drawing. Upon completion of this test the weight loss of the bristles shall be determined. The weight loss of the bristles shall be the original weight of the brush minus the weight of the brush after the test.

* 4.6.3 Performance-chamber cleaning brushes. The brush shall be inserted into the full depth of the chamber, rotated clockwise one (1) revolution and removed the number of times specified in 3.3.2. A 360° movement shall be considered as one revolution. Speed of revolution shall be 15 to 30 revolutions per minute. Chambers used for this test shall be of the type and caliber specified on the applicable brush drawing. Upon completion of this test, the weight loss of the bristles shall be determined as specified in 4.6.2.

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* 4.6.4 Performance-firing pin hole cleaning brushes. The bristle portion of the brush shall be pushed in one continuous motion through the length of the firing-pin hole of the applicable bolt and returned. This movement shall be counted as two strokes. The bristle portion of the brush shall be forced entirely clear of the firing-pin hole at the end of each stroke. Speed of travel for each stroke shall be approximately 2 feet per second. The number of strokes required for each test brush shall be as specified in 3.3.3. Bolts used for this test shall be of the type and caliber specified. Upon completion of this test the weight loss of the bristles shall be determined as specified in 4.6.2.

4.6.5 Nail load. The brush shall be assembled into the Government approved test gage and subjected to the nail load test. The brush is defective if the applied load disengages the nail completely from the adapter.

* 4.6.6 Tensile pull. The brush shall be assembled into the Government approved test gage and subjected to the tensile pull test. The brush is defective if the brush does not meet the requirements specified on the applicable drawing.

4.7 Inspection equipment. Inspection equipment utilized in quality conformance inspection shall be as listed on the applicable Inspection Equipment List (see 6.2).

5. PREPARATION FOR DELIVERY

* 5.1 Pilot pack. The pilot pack shall consist of a brush or brushes as required by and packaged in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

* 5.2 Preservation, packaging, packing and marking. Brushes shall be preserved, packaged, packed and marked in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2).

6. NOTES

* 6.1 Intended use. This specification is intended to cover, but not be limited to, procurement of the following brushes:

Drawing	Nomenclature
C5504036	Brush, Cleaning, Cal. .45, M5
C5504037	Brush, Cleaning, Cal. .50, M4, Bore
C5564174	Brush, Cleaning, Cal. .30
C5564179	Brush, Cleaning, Cal. .22, M3

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Drawing	Nomenclature
D6528362	Brush, Chamber Cleaning, M1
C7162132	Brush, Cleaning, Cal. .38
B7162702	Brush, Cleaning, Firing Pin Hole, Cal. .50
B7225087	Brush, Bore, M25, 20MM
B7790452	Brush, Chamber Cleaning, 7.62MM
B7790582	Brush, Chamber Cleaning, Cal. .30
B7790617	Brush, 40MM
B7790665	Brush Assembly, 40MM
B7790737	Brush, Cleaning, Small Arms: Chamber (Cal. .50)
B7791063	Brush, Chamber Cleaning, 20MM
C8407954	Brush, Chamber Cleaning (Cal. .50 - M6 and M3)
C8432358	Brush, Chamber Cleaning (Rifle, M16 and M16A1)
B11010305	Brush Assembly, 40MM
C11686340	Brush, Cleaning, Small Arms: (Rifle 5.56MM, M16)

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of applicable drawings and specifications pertinent to the brushes, showing applicable revision dates.
c. Requirements for submission of first article and shipping instructions for first article and pilot packages (see 3.1, 4.4 and 5.1).
d. Applicable packaging data sheet (see 6.3).
e. Selection of applicable levels of packaging and packing.
f. Master Lists of Inspection and Test Equipment (see 4.7).
g. Government property to be furnished to the contractor and responsibilities therefore (see 4.5.4.1).

* 6.3 The following packaging data sheets are applicable to the brushes covered by this specification as specified in 6.1.

P5504036	Packaging Data Sheet, Brush, Cleaning, Cal. .45, M5
P5504037	Packaging Data Sheet, Brush, Cleaning, Cal. .50, M4, Bore
P5564174	Packaging Data Sheet, Brush, Cleaning, Cal. .30
P5564179	Packaging Data Sheet, Brush, Cleaning, Cal. .22, M3
P6528362	Packaging Data Sheet, Brush, Chamber Cleaning, M1
P7162132	Packaging Data Sheet, Brush, Cleaning, Cal. .38
P7162702	Packaging Data Sheet, Brush, Cleaning, Firing Pin Hole, Cal. .50
P7225087	Packaging Data Sheet, Brush, Bore, M25, 20MM

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P7790452	Packaging Data Sheet, Brush, Chamber Cleaning, 7.62MM
P7790582	Packaging Data Sheet, Brush, Chamber Cleaning, Cal. .30
P7790665	Packaging Data Sheet, Brush Assembly, 40MM
P7790737	Packaging Data Sheet, Brush, Cleaning, Small Arms, Chamber (Cal. .50)
P11010305	Packaging Data Sheet, Brush Assembly, 40MM
P7791063	Packaging Data Sheet, Brush, Chamber Cleaning, 20MM
P8407954	Packaging Data Sheet, Brush, Chamber Cleaning, (Cal. .50 - (M6 and M3)
P8432358	Packaging Data Sheet, Brush, Chamber Cleaning, (Rifle, M16 and M16E1)
P11686340	Packaging Data Sheet, Brush, Cleaning, Small Arms: (Rifle, 5.56MM, M16)

6.4 Management control system requirements. When warranted, the contract should specify the application of MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.5 Performance tests. The Government installation or activity will conduct the quality conformance performance test when it is not feasible to furnish the contractor the barrels or bolts needed for the tests.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Air Force - 84
Navy - OS

Review Activities:
Air Force - 84
Navy - OS

User Activities:
Air Force - 26
Navy - MC

Preparing Activity:
Army - WC

Project Number:
1005-0465

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